Zerun Niu

MPhil Student in Computer Science

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Summary

MPhil student at The University of Sydney working on federated learning, semantic communication, and efficient AI systems. Research interests include distributed optimization, edge intelligence, parameter-efficient tuning, GenAI systems, and ML systems for resource-constrained environments.

Education

Master of Philosophy — **The University of Sydney**, Computer Science — Sydney, Australia — 2025 – present

- Supervisor: Nguyen Tran
- Research focus: Federated Learning, Semantic Communication, and Efficient AI Systems

Bachelor of Advanced Computing — **The University of Sydney**, Data Science — Sydney, Australia — 2025

- Graduated with Distinction.
- Final years WAM: 77/100
- Relevant coursework: Machine Learning, Deep Learning, Natural Language Processing, Distributed Systems

Experience

The University of Sydney, Casual Academic Tutor — Sydney, Australia — Feb 2026 – present — 5 months

Tutorial delivery and student learning support for university coursework.

- Delivered weekly tutorials, class activities, small-group discussions, and Q&A on course content and university learning platforms.
- Supported students in developing academic communication, teamwork, and professionalism skills through guided exercises and feedback.
- Created an inclusive tutorial environment to help students transition into university study and participate confidently.
- Coordinated with unit staff and followed teaching guides to run tutorials consistently and effectively.

DUAL Group, The University of Sydney, Research Assistant — Sydney, Australia — Apr 2024 – present — 2 years 3 months

Research engineering and experimentation for distributed learning and efficient AI systems.

- Built reproducible training and evaluation pipelines with configs, seeds, logging, and automated reporting.
- Supported submission and revision cycles by implementing baselines, targeted ablations, and publication-quality figures and tables.
- Addressed reviewer feedback through additional experiments and clarity edits in method and experiment sections.
- Coordinated experiment ownership, timelines, and artifact handoff across code, plots, and tables to reduce duplicated effort.
- Designed and deployed the official DUAL Group website for members, publications, and research projects.

Publications

Distributionally Robust Wireless Semantic Communication with Large AI Models 2025
L. T. Le, S. H. Wanasekara, *Zerun Niu*, et al.
10.48550/arXiv.2506.03167 (IEEE Journal on Selected Areas in Communications)

Federated Deep Equilibrium Learning over Resource-Constrained Edge Networks 2025
L. T. Le, *Zerun Niu*, T. D. Nguyen, et al.
(Submitted to IEEE Internet of Things Journal)

Projects

Adaptive Low-Rank Optimization for Federated Learning 2025
Undergraduate thesis integrating DyLoRA into a federated learning pipeline.
- Reduced communication cost by 80% without loss of accuracy.
- Improved training stability across clients with reproducible PyTorch experiments.
- Result: 85/100, High Distinction.

DUAL Group Website 2024-2025
Designed and deployed the official website for the Distributed compUting, optimizAtion, and Learning group.
- Introduced group members, publications, and research projects.
- Website: https://dual-grp.github.io/website-dual/

Teaching

Casual Academic Tutor 2026-02 to present
University of Sydney tutorial delivery and student learning support.
- Tutorial delivery: weekly tutorials, class activities, small-group discussions, and student Q&A.
- Subjects: Algorithms and Data Structures, Computer Systems, Machine Learning, Deep Learning, NLP, Data Science Foundations, Distributed Systems.
- Assessment: assignments, coding practicals, project reports, oral demos, rubric-aligned marking.
- Student support: consultations, Ed discussion, code reviews, debugging practices, academic integrity and citation support.

Skills

GenAI and Multimodal: LLM/VLM orchestration, agents, tool use, multimodal adapters, LoRA, alignment, safety guardrails, jailbreak defense, RAG, automated evaluation, and human-in-the-loop evaluation.

Efficient Tuning: LoRA, DyLoRA, PEFT workflows, quantization, distillation, and efficient adaptation under constrained compute budgets.

ML Engineering: PyTorch, Hugging Face Transformers, PEFT, scikit-learn, reproducible training pipelines, experiment tracking, modular pipeline design, and API development.

Distributed and Data Systems: Ray, Dask, Databricks, Spark, Hadoop, Airflow, FastAPI, latency and performance optimization, tracing, and observability.

Programming: Python, SQL, Java, C/C++, Git/GitHub, Linux, Jupyter, PostgreSQL, MySQL, MongoDB, Pandas, and NumPy.

Awards

Undergraduate Thesis High Distinction 2025
Adaptive Low-Rank Optimization for Federated Learning, score 85/100.